

Unit-3 Revision: Exploratory Data Analysis

CSE273: Foundation of Machine Learning

Outline

- 1 Basics of EDA
- 2 Univariate Analysis
- 3 Bivariate Analysis
- 4 Correlation & Distributions
- 5 Advanced EDA

Exploratory Data Analysis (EDA)

- **Definition:** EDA is an approach to analyzing datasets to summarize their main characteristics, often using statistical graphics and other data visualization methods.
- **Goal:** To understand the data, discover patterns, spot anomalies, frame hypotheses, and check assumptions before formal modeling.
- **Key Components:**
 - Understanding Data Types
 - Univariate Analysis (One variable)
 - Bivariate/Multivariate Analysis (Two or more variables)
 - Identifying Missing Values and Outliers

Types of Data

- **Numerical (Quantitative):** Data that can be measured and expressed as numbers.
 - *Continuous:* Infinite number of possible values (e.g., Height, Weight, Temperature).
 - *Discrete:* Countable number of distinct values (e.g., Number of children, Cars owned).
- **Categorical (Qualitative):** Data that describes categories or groups.
 - *Nominal:* Categories with no inherent order (e.g., Colors, Gender, Blood Type).
 - *Ordinal:* Categories with a meaningful order or ranking (e.g., Education level, Customer satisfaction ratings).

Loading Data using Pandas

- The first step in EDA is loading the dataset into a usable format, typically a Pandas DataFrame.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Load dataset
df = pd.read_csv('data.csv')
```

```
# Initial Inspection
display(df.head())      # View first few rows
display(df.info())      # Check data types and nulls
display(df.describe())  # Statistical summary of numerical columns
```

Univariate Analysis: Histograms & Count Plots

Histogram (Numerical)

- Visualizes the frequency distribution of a continuous variable by grouping values into "bins".

```
# Seaborn Histogram
sns.histplot(df['Age'],
             bins=20,
             kde=True)
plt.title('Age Distribution')
plt.show()
```

Count Plot (Categorical)

- Shows the counts of observations in each categorical bin using bars.

```
# Seaborn Count Plot
sns.countplot(x='Gender',
              data=df)
plt.title('Gender Count')
plt.show()
```

Univariate Analysis: Box Plot

- **Box Plot:** A standardized way of displaying the distribution of data based on a five-number summary: Minimum, First Quartile (Q1), Median (Q2), Third Quartile (Q3), and Maximum.
- **Primary Use:** Highly effective for identifying **outliers** and understanding the spread/skewness of the data.

Creating a Box Plot

```
sns.boxplot(x=df['Salary'])  
plt.title('Salary Box Plot')  
plt.show()
```

- *Interquartile Range (IQR):* $Q3 - Q1$. Outliers are typically points falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

Bivariate Analysis: Scatter Line Plots

Scatter Plot

- Shows relationship between two *numerical* variables.
- Helps identify correlation and clusters.

```
sns.scatterplot(x='Age',  
                y='Income',  
                data=df)  
  
plt.show()
```

Line Plot

- Shows trends in numerical data over a continuous interval (often time).

```
sns.lineplot(x='Year',  
             y='Sales',  
             data=df)  
  
plt.show()
```


Bivariate Analysis: Category vs Numeric (Violin Plots)

- **Violin Plot:** Combines a box plot with a kernel density plot (KDE).
- **Purpose:** Excellent for comparing the distribution of a *numerical* variable across different levels of a *categorical* variable.
- It shows the median, IQR, and the full shape of the data distribution (including multiple peaks).

```
# Comparing Salary distribution across different Departments
sns.violinplot(x='Department',
               y='Salary',
               data=df)

plt.title('Salary Distribution by Department')
plt.show()
```

Correlation Analysis using Heatmaps

- **Correlation:** A statistical measure that expresses the extent to which two numerical variables change together (ranging from -1 to 1).
- **Heatmap:** A 2D visual representation of data where values are encoded in colors. Perfect for visualizing correlation matrices.

```
# Calculate correlation matrix
corr_matrix = df.corr()

# Plot heatmap
sns.heatmap(corr_matrix,
            annot=True,      # Show correlation values
            cmap='coolwarm', # Color map
            fmt='.2f')       # Format to 2 decimal places

plt.show()
```

Multicollinearity

- **Definition:** A phenomenon where two or more predictor variables (features) in a multiple regression model are highly correlated.
- **Impact:** It makes it difficult to determine the individual effect of each feature on the target variable, leading to unstable coefficients and reduced model interpretability.
- **Detection:**
 - Correlation Matrices (Heatmaps).
 - **Variance Inflation Factor (VIF):** Measures how much the variance of an estimated regression coefficient increases if predictors are correlated. $VIF > 5$ or 10 indicates high multicollinearity.
- **Solution:** Remove highly correlated features or use dimensionality reduction (like PCA).

Distribution Analysis: Skewness and Kurtosis

- **Skewness:** A measure of the asymmetry of the probability distribution.
 - *Positive (Right) Skew:* Tail is longer on the right. $\text{Mean} > \text{Median}$.
 - *Negative (Left) Skew:* Tail is longer on the left. $\text{Mean} < \text{Median}$.
 - *Zero Skew:* Symmetrical (Normal Distribution).
- **Kurtosis:** A measure of the "tailedness" of the distribution (how heavy or light the tails are compared to a normal distribution).
 - *Leptokurtic (High Kurtosis):* Heavy tails, more outliers.
 - *Platykurtic (Low Kurtosis):* Light tails, fewer outliers.
 - *Mesokurtic:* Normal distribution.

Detecting Patterns, Anomalies and Trends

- **Patterns:** Regular, repeating structures in data (e.g., seasonality in time series). Detected using line plots and scatter plots.
- **Trends:** Long-term upward or downward movements in the data. Evaluated using moving averages or regression lines.
- **Anomalies (Outliers):** Data points that deviate significantly from other observations.
 - *Detection Methods:* Visual inspection (Box plots, Scatter plots), Statistical methods (Z-score, IQR method), or ML methods (Isolation Forests).

EDA Workflow Integration

- How EDA fits into the standard Machine Learning Pipeline:
- ① **Data Acquisition:** Gather data.
 - ② **Data Cleaning:** Handle missing values and duplicates.
 - ③ **EDA (Iterative Step):**
 - Univariate and Bivariate analysis.
 - Understand feature distributions and relationships.
 - Identify anomalies and necessary transformations.
 - ④ **Feature Engineering:** Create new features or transform existing ones (e.g., handling skewness) based on EDA insights.
 - ⑤ **Model Training:** Proceed to model building with a deep understanding of the dataset.

End of Unit-3 Revision

Any Questions?